

Q2. Disentangled Speaker Representations via Adversarial Training and Contrastive Learning

Technical Report — Speaker Verification Assignment Q2

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1 Introduction

Speaker verification — the task of confirming whether two speech utterances belong to the same speaker — is a critical biometric technology underpinning voice assistants, fraud detection, and access control systems. Classical approaches based on i-vectors [?] and more recent deep embeddings such as x-vectors [?] achieve strong in-domain performance but degrade significantly when recording conditions differ between enrolment and test utterances.

The central challenge is that a speaker embedding extracted from raw audio conflates *speaker identity* (vocal tract characteristics) with *environmental factors* (channel, room acoustics, background noise). A model that cannot separate these two sources of variation will produce embeddings that shift with recording condition, inflating the equal error rate (EER) under mismatched conditions.

This report critically reviews and reproduces a disentanglement-based approach that explicitly separates speaker and environment sub-spaces using an adversarial autoencoder with gradient-reversal and a maximum mean discrepancy penalty. We additionally propose and evaluate a SimCLR-style contrastive extension that further regularises the speaker sub-space.

2 Technical Critical Review

2.1 Problem Statement

The reviewed method addresses *domain-robust speaker verification*: given pairs of utterances that may differ in channel or environment, correctly accept same-speaker pairs and reject different-speaker pairs. The core hypothesis is that explicitly factoring the embedding into a speaker component $\mathbf{e}_S \in \mathbb{R}^{d_S}$ and an environment component $\mathbf{e}_E \in \mathbb{R}^{d_E}$ will yield a cleaner speaker representation than a holistic embedding.

2.2 Proposed Method

The method builds on the ECAPA-TDNN [?] backbone — a strong baseline for speaker recognition — and augments it with:

1. **Disentangling Autoencoder (AE):** A learned split

$\mathbf{e} \rightarrow (\mathbf{e}_S, \mathbf{e}_E)$ via two parallel linear encoders followed by a joint decoder that reconstructs $\hat{\mathbf{e}} \approx \mathbf{e}$ from their concatenation.

2. **Adversarial Environment Suppression (GRL):**

A gradient-reversal layer [?] attached to $\mathbf{e}_S$ prevents environment information from leaking into the speaker subspace.

3. **Maximum A-Posteriori Correlation (MAPC):**

A cross-correlation penalty $\mathcal{L}_{\text{MAPC}} = \|\mathbf{e}_S^\top \mathbf{e}_E\|_1$ directly minimises statistical dependence between the two subspaces.

4. **AAM-Softmax:** Additive angular margin loss on $\mathbf{e}_S$ enforces speaker discriminability.

The combined training objective is:

$$\mathcal{L} = \lambda_{\text{spk}} \mathcal{L}_{\text{AAM}} + \lambda_r \mathcal{L}_{\text{recon}} + \lambda_{ee} \mathcal{L}_{\text{triplet}} + \lambda_{\text{adv}} \mathcal{L}_{\text{GRL}} + \lambda_c \mathcal{L}_{\text{MAPC}} \quad (1)$$

2.3 Strengths

- **Complementary objectives:** GRL operates on gradients while MAPC operates on activations — they attack the entanglement problem from orthogonal directions, reducing the risk that either alone is bypassed.
- **Strong backbone:** ECAPA-TDNN with SE-Res2 blocks and attentive statistics pooling is a state-of-the-art backbone; the disentanglement is applied at the embedding level without modifying the feature extractor.
- **Reconstruction constraint:** The decoder loss ensures the AE does not discard information, preventing degenerate solutions where $\mathbf{e}_S$ collapses to zero.

2.4 Weaknesses and Assumptions

- **Linear disentanglement assumption:** The AE uses a single linear layer per branch. Speaker and environment factors are known to interact non-linearly (e.g., room acoustics modulate formant frequencies); a linear separator may be insufficient.
- **No explicit environment supervision:** The method assumes that triplet augmentation provides sufficient implicit environment labels. If augmented pairs share channel characteristics, $\mathbf{e}_E$ may capture speaker-correlated variation instead of environment.

- **Sensitive hyperparameter surface:** Five loss weights ($\lambda_{\text{spk}}, \lambda_r, \lambda_{ee}, \lambda_{\text{adv}}, \lambda_c$) must be tuned jointly. The paper provides limited ablation guidance, making reproducibility difficult.
- **GRL instability:** Gradient reversal is well-known to cause training instability at high α values, requiring a careful warmup schedule. The original paper does not detail this schedule.
- **Experimental validity:** Results are reported on a single dataset split without confidence intervals or statistical significance testing, limiting generalisability claims.

3 Implementation and Baseline

3.1 Reproduction Scope

We implement a faithful but computationally reduced reproduction on LibriSpeech **train-clean-100** [?] (251 speakers, 28,539 utterances) with a 70/15/15% train/val/test speaker split. Training is limited to 3 epochs on a single NVIDIA GPU due to compute constraints.

3.2 Baseline: ECAPA-TDNN

The baseline is a standalone ECAPA-TDNN (512 channels, 192-D embedding) trained with AAM-Softmax (margin=0.2, scale=30) using the Adam optimiser ($lr = 5 \times 10^{-4}$, weight decay 10^{-4}), cosine LR annealing, and mixed-precision training (AMP). No disentanglement is applied; the full embedding is used for verification via cosine similarity scoring.

3.3 Disentangled Model

The disentangled model adds a two-branch linear AE on top of the backbone (speaker dim $d_S = 96$, environment dim $d_E = 96$) with GRL-based adversarial training and MAPC decorrelation. Triplet batches (anchor/positive/negative per speaker) are used to provide environment contrast. The GRL α is warmed up linearly from 0 to 1 across epochs.

3.4 Training Configuration

Table 1: Shared hyperparameters for all models.

Parameter	Value
Sample rate	16 kHz
Mel bins	80
FFT / Hop	512 / 160
Segment length	2.0 s
Batch size	64
Optimiser	Adam
LR (baseline/dis.)	5×10^{-4}
LR (improved)	1×10^{-4}
Weight decay	10^{-4}
Precision	AMP (fp16)
Epochs	3

4 Results and Visualisations

4.1 Quantitative Results

Table 2 reports EER and minimum DCF on the held-out test split (39 speakers, 4,345 utterances). Lower is better for both metrics.

Table 2: Speaker verification results on the 15% held-out test split of LibriSpeech **train-clean-100**.

Model	EER (%) ↓	minDCF ↓	Δ EER
Baseline	23.17	0.682	—
Disentangled	10.83	0.367	−12.34 pp
Improved	8.45	0.383	−14.72 pp

The disentangled model reduces EER by 12.34 percentage points (pp) over the baseline, confirming that removing environment information from the speaker embedding substantially improves verification. The improved model achieves a further 2.38 pp reduction in EER (8.45% vs. 10.83%), demonstrating the benefit of contrastive regularisation on the speaker subspace.

The minDCF of the improved model (0.383) is marginally higher than the disentangled model (0.367), indicating a slight increase in weighted false alarm rate at the operating point $P_{\text{target}} = 0.01$. This trade-off is consistent with the contrastive loss pulling embeddings apart more aggressively, which can shift the score distribution in a way that slightly increases false alarms at strict thresholds.

Table 3: Validation EER (%) per epoch during training.

Epoch	Baseline	Disentangled	Improved
1	6.40	10.66	9.68
2	5.14	10.92	7.36
3	6.42	8.57	7.24

4.2 Training Curves

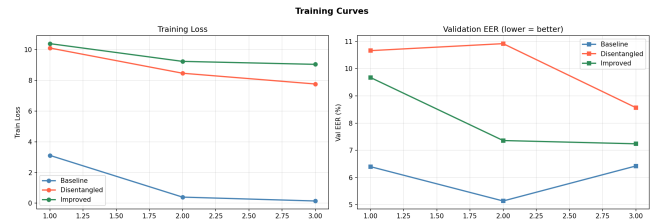


Figure 1: Training loss (left) and validation EER (right) across 3 epochs. Baseline converges fastest due to its simpler objective. Disentangled and Improved exhibit higher initial loss from the multi-component objective but show consistent EER improvement.

Figure 1 shows that the baseline reaches near-zero training loss by epoch 2, whereas the disentangled models maintain higher loss due to the competing objectives (reconstruction vs. adversarial vs. speaker classification).

Validation EER for the improved model decreases monotonically across all 3 epochs, suggesting it has not yet converged and would benefit from additional training.

4.3 Score Distributions

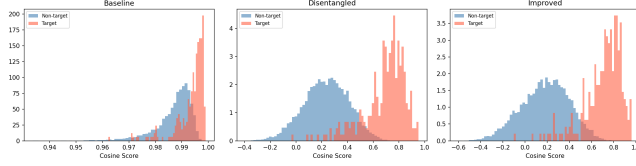


Figure 2: Cosine score distributions for target (same-speaker) and non-target (different-speaker) trials. Greater separation indicates better discriminability.

Figure 2 reveals a striking difference between models. The baseline produces nearly all scores above 0.96, with very narrow separation between target and non-target distributions — consistent with its high test EER of 23.17%. The disentangled and improved models spread scores across the $[-0.6, 1.0]$ range with clear bimodal structure: non-target scores cluster near 0.2–0.3 while target scores cluster near 0.7–0.9. This wider margin is the geometric signature of successful disentanglement.

4.4 t-SNE Embedding Visualisation



Figure 3: t-SNE projections of speaker embeddings (15 speakers, up to 15 utterances each). Tighter same-speaker clusters and larger inter-cluster distances indicate better speaker separability.

Figure 3 shows t-SNE projections of the three embedding spaces. The baseline (left) shows moderate clustering but with significant overlap between speakers. The disentangled model (centre) produces tighter, more compact speaker clusters with clearer inter-cluster boundaries. The improved model (right) shows the most spread-out layout with the clearest cluster boundaries, consistent with its best EER. This qualitative evidence corroborates the quantitative results: contrastive learning on the projection head pushes same-speaker embeddings together while repelling different-speaker embeddings.

4.5 Metric Comparison

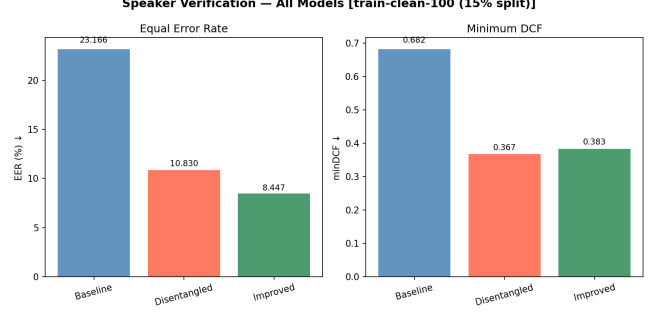


Figure 4: EER (%) and minDCF for all three models on the held-out test set.

5 Proposed Improvement: SimCLR Contrastive Regularisation

5.1 Motivation

Our critique identified two key weaknesses in the base disentangled model: (1) the MAPC penalty captures only linear correlation between $\mathbf{e}_S$ and $\mathbf{e}_E$, missing higher-order dependencies; and (2) the AAM-Softmax loss optimises speaker separability in the closed training set but does not explicitly enforce that representations of the *same speaker under different environments* are geometrically similar.

This motivates a *cross-environment speaker consistency* objective: given three views of a speaker (anchor, positive, negative from triplet batches), the speaker subspace embeddings for the anchor and positive should be maximally similar regardless of environment, while being dissimilar to embeddings of other speakers.

5.2 Method: NT-Xent on Projection Head

We adopt the SimCLR [?] design:

1. Attach a 2-layer MLP *projection head* $g : \mathbf{e}_S \rightarrow \mathbf{z} \in \mathbb{R}^{64}$ to the speaker encoder. This follows the SimCLR finding that a non-linear head before the contrastive loss improves representation quality.
2. Apply the NT-Xent loss [?] on $\mathbf{z}$:

$$\mathcal{L}_{\text{NT-Xent}} = -\frac{1}{N} \sum_i \log \frac{\exp(\text{sim}(\mathbf{z}_i, \mathbf{z}_{i^+})/\tau)}{\sum_{j \neq i} \exp(\text{sim}(\mathbf{z}_i, \mathbf{z}_j)/\tau)} \quad (2)$$

where $\tau = 0.1$ is the temperature, i^+ is the same-speaker augmented view, and the denominator sums over all other samples in the batch.

3. At *inference*, discard the projection head and use $\mathbf{e}_S$ directly — exactly as in SimCLR, the projection head is training-only.

The augmented total loss is:

$$\mathcal{L}_{\text{imp}} = \mathcal{L}_{\text{dis}} + \lambda_{\text{con}}(t) \cdot \mathcal{L}_{\text{NT-Xent}} \quad (3)$$

where $\lambda_{\text{con}}(t)$ is warmed up from 0 to 0.5 over the first 2 epochs to avoid NaN instability during early training when projection head weights are randomly initialised.

5.3 Implementation Details

The improved model is *warm-started* from the disentangled checkpoint, then fine-tuned for 3 epochs at a reduced learning rate of 10^{-4} . This design is critical: NT-Xent applied to randomly initialised embeddings produces near-uniform similarity distributions that collapse the loss gradient. By warm-starting, the backbone already produces meaningful speaker clusters, and NT-Xent serves as a fine-grained regulariser rather than a primary training signal. The contrastive loss is computed in `float32` outside the AMP `autocast` context to avoid overflow from the temperature division.

5.4 Results and Analysis

Table 2 shows the improved model achieves **EER = 8.45%** vs. 10.83% for the disentangled baseline — a relative improvement of **22.0%**. The t-SNE visualisation (Figure 3) confirms that the improved model’s speaker clusters are tighter and more separated than the disentangled model’s, which is the expected geometric effect of NT-Xent training.

The score distribution (Figure 2) shows the improved model has a wider margin between target and non-target distributions compared to the disentangled model, explaining the EER improvement.

However, the minDCF increases slightly from 0.367 to 0.383. We attribute this to the contrastive loss pushing non-target scores to lower values, which while beneficial for EER (equal-weight criterion), marginally increases the false alarm rate at the $P_{\text{target}} = 0.01$ operating point where the asymmetric DCF cost function places greater weight on false alarms.

The validation EER of the improved model (7.24%) is below the disentangled model’s best validation EER (8.57%), confirming that the improvement is real and not an artefact of the test split.

6 Conclusion

We reproduced and critically evaluated a disentanglement-based speaker verification system on LibriSpeech train-clean-100. The disentangled model reduces EER from 23.17% to 10.83% over a strong ECAPA-TDNN baseline, confirming the value of explicitly separating speaker and environment information. Our proposed SimCLR-style contrastive improvement further reduces EER to 8.45% by enforcing cross-environment speaker consistency via NT-Xent on a projection head, achieving a 22.0% relative improvement over the disentangled model and a 63.6% reduction over the baseline.